Wall Clutter Suppression Method Based on Joint Dual-Channel Low-Rank and Sparse Decomposition for UAV-Mounted TTW-SAR

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***Abstract*—Uncrewed aerial vehicle (UAV)-mounted through-the-wall synthetic aperture radar (TTW-SAR) is critical for urban building detection, holding significant value in civilian and military applications. However, wall clutter is typically much stronger, severely degrading imaging quality. To address this issue, this article proposes a joint dual-channel low-rank and sparse decomposition (JD-LRSD) method for wall clutter suppression based on UAV-mounted TTW-SAR. In the proposed method, a joint dual-channel low-rank and sparse model is developed, with constraints of the dual-channel similarity and wall clutter distribution. Then, the alternating direction method of multipliers (ADMM) is employed to solve the developed optimization problem. Closed-form solutions of each subproblem are derived as explicit expressions. Finally, numerical simulations and field experiments are carried on a dual-channel UAV TTW-SAR system. The proposed method improves the signal-to-interference-plus-noise ratio (SINR), outperforming comparison methods by more than 9 dB.**

***Index Terms*—Dual channel, low-rank and sparse decomposition (LRSD), through-the-wall radar, uncrewed aerial vehicle (UAV), wall clutter suppression.**

I. Introduction

UNCREWED aerial vehicle (UAV)-mounted through-the-wall synthetic aperture radar (TTW-SAR) plays an essential role in civilian and military applications. Through the penetration perception of urban architectures, UAV-mounted TTW-SAR can provide target information inside buildings for further layout reconstruction [1], [2], target tracking [3], [4], and human motion recognition [5], [6], in which a high-precision image is a prerequisite for these tasks. However, wall clutter is typically much stronger, obscuring target echoes from area of interest and causing low signal-to-interference-plus-noise ratio (SINR).

Wall clutter suppression has received extensive attention and research, including background subtraction, time gating, spatial filtering, subspace projection, low-rank and sparse decomposition (LRSD), and compressed sensing (CS). The background subtraction method required two data collections in obstructed regions with and without targets, and the subtraction was conducted to completely remove the wall clutter [7]. However, it could not be applied to noncooperative scenarios and the direct subtraction would cause residual wall clutter due to the dynamic trajectory of each collection in UAV-mounted TTW-SAR. The time gating method assumed the wall clutter had similar behavior over each pulse repetition time (PRT) and defined a windowing strategy for clutter suppression [8], which was unreliable for scenarios under low SINR conditions.

The spatial filtering method assumed that the width of the wall was much greater than the beamwidth of the antenna and wall clutter was approximately the same across antenna channels [9]. Since targets possessed a limited spatial extent relative to the antenna's beamwidth, their echoes exhibited significant antenna-dependent characteristics. Based on the discrepancy in the spatial-frequency domain between wall and target responses, the wall clutter was sufficiently suppressed. However, its practical applications were constrained, requiring parameter adjustments for different scenarios.

Assuming that wall and target echoes lay in different orthogonal subspaces approximately, the subspace projection method utilized singular value decomposition (SVD) to decompose received echoes into several orthogonal components [10], [11], [12]. The dominant singular values belonged to the wall subspace, while the minor singular values belonged to the noise subspace. Wall clutter suppression was achieved by singular value filtration. The performance of the subspace projection depended on whether the dominant singular value selection could precisely reconstruct wall clutter, which was greatly affected by echo quality.

The LRSD method introduced concepts of low-rank and sparse characteristics, decomposing radar echoes into a low-rank component and a sparse component [13], [14], [15]. Wall responses were approximately antenna-independent and target responses were antenna-dependent. Thus, wall and target responses were reasonable to be modeled as low-rank and sparse matrices, respectively. Since the matrix rank constraint and the ℓ_0-norm constraint constituted an ill-posed nonconvex optimization, the nuclear norm constraint and the ℓ_1-norm constraint were adopted to convert the LRSD problem into a well-posed convex optimization. Some pioneer efforts

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were devoted, including robust principal component analysis (RPCA) and its variant forms [16], [17], [18]. Later, weighted nuclear norm minimization (WNNM) was applied to the low-rank matrix by assigning different penalty weights to singular values for more accurate low-rank approximation [19], [20]. Furthermore, total variation regularization was applied to the sparse matrix to enhance the smoothness of target while maintaining its edge integrity [21], [22], [23]. Bilinear decomposition decomposed the low-rank matrix into the product of two substantially smaller factor matrices [24], [25], [26], allowing the SVD operation to be performed directly on the factor matrices and avoiding the complex computation.

The CS-based method employed an overcomplete dictionary matrix to model signal sparsity and utilized convex relaxation, greedy pursuit, or Bayesian optimization for wall echo reconstruction. The dictionary matrix was typically constructed using the Fourier basis and the discrete prolate spheroidal sequence (DPSS) basis. The Fourier basis was appropriate for point targets [27], [28], while the DPSS basis was preferred for extended targets owing to its band-limited property [29], [30]. The Fourier-based method relied on wall echo modeling and utilized a maximum correlation criterion to substitute wall parameters into the signal model to approximate actual wall responses [31], [32]. Although the band-limited property offered superior energy concentration and avoided spectrum leakage in wall echo reconstruction, the DPSS basis was independent of radar waveform, lacking the discriminability to separate wall and target echoes. Hence, the DPSS-based method relied on extra prior information, such as time gating and energy ratio [33], [34]. Constrained by prior information, the method needed to be well-adjusted for dynamic scenarios, limiting the practical utility.

Nevertheless, conventional CS-based approaches were generally formulated within the single-measurement vector (SMV) framework [35], [36]. Assuming multiple measurements shared the same nonzero sparse support, commonly observed in multichannel images [37], [38], [39], multipolarization modalities [14], [40], [41], and multipath propagations [42], [43], the multiple-measurement vector (MMV) model was further developed for joint sparse recovery. In [39], the group sparsity of multichannel images along the vertical direction was exploited to ensure the structural coherence in the 3-D layout reconstruction. Leveraging the group sparsity of multipolarization modality, wall clutter suppression and image formation were formulated as a regularized least squares problem for high-resolution imaging [14]. In [43], a structure-aware Bayesian CS algorithm was developed with the group sparsity of multipath propagations and clustering sparsity of extended targets for high-resolution imaging. However, the group sparsity merely restricts the interchannel consistency of sparse positions, neglecting the relationship of sparse values.

Based on the inherent duality between image-domain sparsity and data-domain Hankel rank, structured low-rank matrix recovery was widely investigated for sparse-sampling radar imaging [44], [45]. The twofold Hankel structure is constructed to enhance the low-rank property and the alternating direction method of multipliers (ADMM) is adopted with Hankel matrix factorization for ISAR imaging [46]. To alleviate the dependence on specific sparse-sampling patterns, the structured low-rank constraint based on Hankel formulation is further combined with an analytical sparsity model to accommodate diverse sampling scenarios [47]. Then, the joint sparse constraint of the MMV is introduced into structured low-rank matrix recovery with the designed multichannel Hankel matrix for multichannel radar imaging in [48]. However, the structured low-rank technique designed for target imaging in remote sensing lacks the adaptability for wall clutter suppression in TTW scenarios.

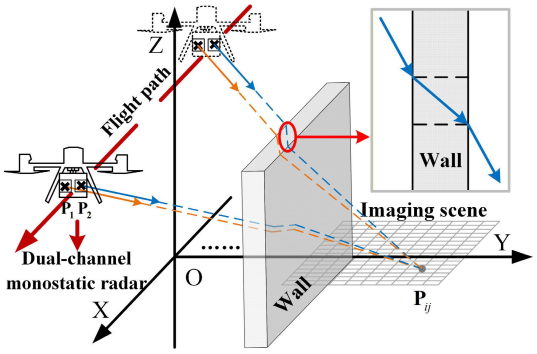


Fig. 1. Geometric model of dual-channel UAV-mounted TTW-SAR.

In this article, a joint dual-channel LRSD (JD-LRSD) method is proposed for wall clutter suppression based on UAV-mounted TTW-SAR. The method is established on a monostatic dual-channel TTW radar system with the main contributions summarized as follows.

1) Through the analysis of the wall clutter with motion errors under the small channel interval assumption, dual-channel similarity is designed to exploit the interchannel consistency of sparse values.
2) Exploiting the spatial extended clustering property of wall structure, wall clutter distribution is formulated for adaptively focusing on wall clutter regions.
3) The joint dual-channel low-rank and sparse model is established and integrated with dual-channel similarity and wall clutter distribution constraints, whose closed-form solutions for subproblems are derived for efficient optimization by the ADMM algorithm.

This article is organized as follows. Section II introduces the established geometric and signal models of dual-channel UAV-mounted TTW-SAR. Section III constructs a joint dual-channel low-rank and sparse model and proposes an ADMM-based wall clutter suppression method. Section IV presents both numerical simulations and field experiments to verify the wall clutter suppression performance of the proposed method. The conclusion is drawn in Section V.

II. Geometric and Signal Model

As shown in Fig. 1, a monostatic dual-channel TTW radar system is installed on the UAV platform, performing penetration perception for targets behind the wall. The black dotted line is the flight path, which is parallel to the wall surface. The geometric coordinate system is established as O–*XYZ*, where the *X*-, *Y*-, and *Z*-axes denote the azimuth, range, and

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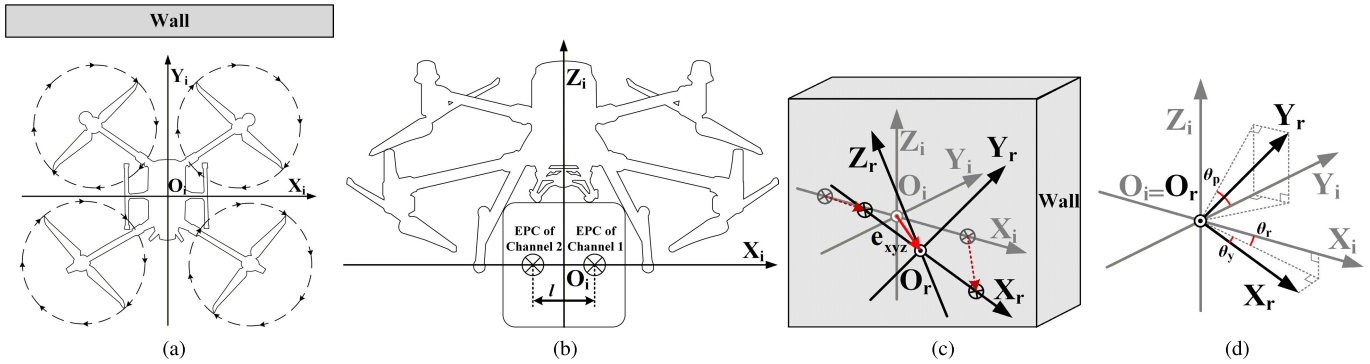


Fig. 2. Array-body coordinate system. (a) and (b) Top view and front view of the ideal array-body coordinate system. (c) Practical array-body coordinate system. (d) Attitude angle transformation.

height directions. The X-axis is parallel to the wall, while the Y-axis is perpendicular to the wall. Simplifying the array configuration with the equivalent phase center (EPC), the dual-channel EPCs are denoted as $\mathbf{P_1} = [X(t_a) + l/2, Y(t_a), Z(t_a)]^{\mathrm{T}}$ and $\mathbf{P_2} = [X(t_a) - l/2, Y(t_a), Z(t_a)]^{\mathrm{T}}$, where t_a is the slow time, superscript T denotes the matrix transpose, and l denotes the channel interval between EPCs.

Similar to the geometric coordinate system O–XYZ, the ideal array-body coordinate system of the UAV platform is established as $\mathrm{O_i}$–$X_iY_iZ_i$ in Fig. 2(a), where the origin point $\mathrm{O_i}$ is the central point between the two channels. The $\mathrm{X_i}$-, $\mathrm{Y_i}$-, and $\mathrm{Z_i}$-axes are the azimuth, range, and height directions. In the coordinate system $\mathrm{O_i}$–$X_iY_iZ_i$, the ideal dual-channel EPCs are denoted as $\mathbf{p_1} = [l/2, 0, 0]^{\mathrm{T}}$ and $\mathbf{p_2} = [-l/2, 0, 0]^{\mathrm{T}}$.

Due to atmospheric turbulence, the UAV platform suffers from undetermined motion errors, causing the deviation from the presupposed ideal flight path. The array-body coordinate system is transformed from $\mathrm{O_i - X_iY_iZ_i}$ to $\mathrm{O_r - X_rY_rZ_r}$ in Fig. 2(c). The motion errors are divided into two independent categories: translational errors and attitude errors. The translational errors denote the deviation of the origin point from $\mathrm{O_i}$ to $\mathrm{O_r}$, drawn as the red solid arrow and recorded as $\mathbf{e_{xyz}} = [\Delta x(t_a), \Delta y(t_a), \Delta z(t_a)]^{\mathrm{T}}$. The attitude errors denote the rotation of the origin coordinate system including the pitch angle $\theta_{\mathrm{p}}(t_a)$, roll angle $\theta_{\mathrm{r}}(t_a)$, and yaw angle $\theta_{\mathrm{y}}(t_a)$ in Fig. 2(d). Affected by motion errors, the practical dual-channel EPCs are denoted as

$$\hat{\mathbf{p}}_\mathbf{i} = \mathbf{H}_{\mathrm{y}}\mathbf{H}_{\mathrm{r}}\mathbf{H}_{\mathrm{p}}\mathbf{p}_\mathbf{i} + \mathbf{e}_{xyz} \tag{1}$$

where i denotes the channel number and $\mathbf{H}_{\mathrm{y}}, \mathbf{H}_{\mathrm{r}}, \mathbf{H}_{\mathrm{p}}$ are transformation matrices of the euler angles $\theta_{\mathrm{y}}(t_a), \theta_{\mathrm{r}}(t_a), \theta_{\mathrm{p}}(t_a)$

$$\mathbf{H}_{\mathrm{y}} = \begin{bmatrix} \cos\theta_{\mathrm{y}}(t_a) & -\sin\theta_{\mathrm{y}}(t_a) & 0 \\ \sin\theta_{\mathrm{y}}(t_a) & \cos\theta_{\mathrm{y}}(t_a) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2a}$$

$$\mathbf{H}_{\mathrm{r}} = \begin{bmatrix} \cos\theta_{\mathrm{r}}(t_a) & 0 & \sin\theta_{\mathrm{r}}(t_a) \\ 0 & 1 & 0 \\ -\sin\theta_{\mathrm{r}}(t_a) & 0 & \cos\theta_{\mathrm{r}}(t_a) \end{bmatrix} \tag{2b}$$

$$\mathbf{H}_{\mathrm{p}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta_{\mathrm{p}}(t_a) & -\sin\theta_{\mathrm{p}}(t_a) \\ 0 & \sin\theta_{\mathrm{p}}(t_a) & \cos\theta_{\mathrm{p}}(t_a) \end{bmatrix}. \tag{2c}$$

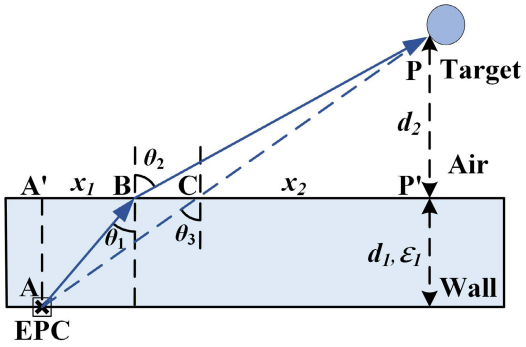


Fig. 3. Geometrical refraction approximation based on the equivalent propagation path from the EPC to the target.

Hence, taking motion errors into consideration, the dual-channel EPCs in the geometric coordinate system O–XYZ are redenoted as

$$\hat{\mathbf{P}}_i = \mathbf{P_i} + \hat{\mathbf{p}}_\mathbf{i} - \mathbf{p_i} = \left[\hat{X}_i(t_a), \hat{Y}_i(t_a), \hat{Z}_i(t_a)\right]^{\mathrm{T}}. \tag{3}$$

In TTW-SAR scenarios, due to the permittivity discrepancy between air and wall, the electromagnetic waves would perform refraction when traveling through the air–wall interface as shown in Fig. 1. Assuming that the wall is homogeneous and much larger than the antenna beamwidth, wall thickness and relative permittivity are denoted as d_1, ε_r. Owing to the symmetry between the radar-to-wall path and the wall-to-target path, the overall propagation path is equivalent to the path drawn in Fig. 3. Considering that the two-way propagation paths between the EPC and the target are equivalent, the one-way propagation path from the EPC to the target is depicted for illustration. Snell's law, along with its approximate forms, provides a foundation for estimating refraction effect and enabling accurate propagation [49], [50], [51]. As illustrated in Fig. 3, a refraction approximation method is employed [50], which substitutes refraction angle θ_2 with θ_3

$$\varepsilon_r = \frac{\cot^2\theta_1 + 1}{\cot^2\theta_3 + 1} = \frac{d_1^2/x_1^2 + 1}{(d_1 + d_2)^2/(x_1 + x_2)^2 + 1} \tag{4}$$

where x_1, x_2, d_1, and d_2 are the lengths of AB', BP', AA', and PP'.

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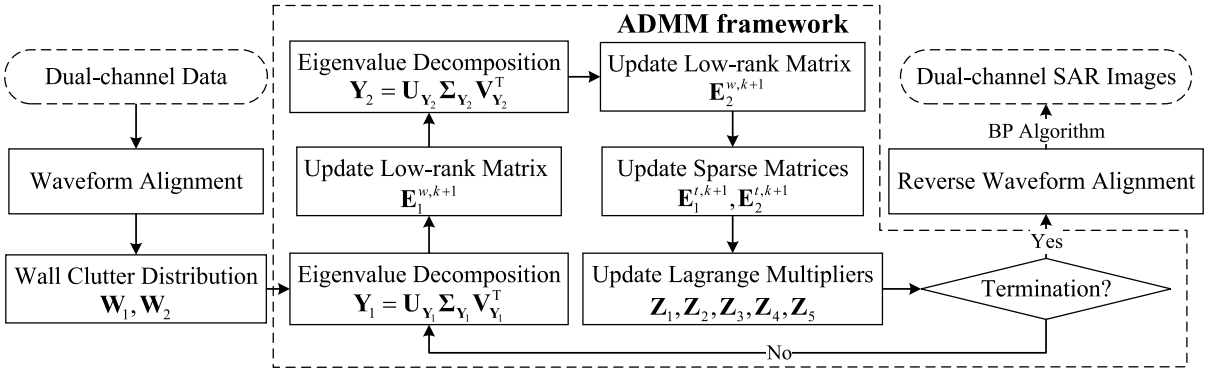


Fig. 4. Flowchart of the proposed JD-LRSD method.

Assuming that an arbitrary point target is located at (x_0, y_0, z_0) behind the wall, the equivalent slant range from the ith EPC to the point target is constructed as

$$R_i(t_a, x_0, y_0) = \sqrt{\varepsilon_r}\sqrt{x_1^2 + d_1^2} + \sqrt{x_2^2 + d_2^2} \tag{5}$$

where d_2, x_1, and x_2 are derived from refraction approximation

$$d_2 = \left|\hat{Y}_i(t_a) - y_0\right| - d_1 \tag{6a}$$

$$x_1 = d_1\left[\frac{\varepsilon_r\left(\hat{Y}_i(t_a) - y_0\right)^2}{\left(\hat{X}_i(t_a) - x_0\right)^2 + \hat{Z}_i^2(t_a)} + \varepsilon_r - 1\right]^{-0.5} \tag{6b}$$

$$x_2 = \sqrt{\left(\hat{X}_i(t_a) - x_0\right)^2 + \hat{Z}_i^2(t_a)} - x_1. \tag{6c}$$

Adopting a linear frequency-modulated continuous wave (LFMCW) radar system, after demodulation and range compression, the dual-channel received signals are expressed as

$$s_i(t_r, t_a) = \mathrm{sinc}\left[B\left(t_r - \frac{2R_i(t_a, x_0, y_0)}{c}\right)\right] \times \exp\left(-j\frac{4\pi f_c}{c}R_i(t_a, x_0, y_0)\right) \tag{7}$$

where B is the signal bandwidth, f_c is the carrier frequency, c is the speed of light, and t_r denotes the fast time.

Since wall clutter is approximately the same across PRTs and target echoes only occupy a small area of interest with most elements being zero, the wall clutter and target echoes exhibit low-rank and sparse characteristics, respectively. The conventional principal component pursuit [16] decomposes the data matrix into two terms, requiring strict low-rank and sparsity without noise interference, while stable principal component pursuit [52] provides a three-term stable decomposition, explicitly separating target echoes and noise without mutual interference. Thus, the amplitude matrix of (7) is decomposed into three terms: $\mathbf{E} = \mathbf{E}^w + \mathbf{E}^t + \mathbf{N} \in \mathbb{R}^{m\times n}$, where $\mathbf{E}^w, \mathbf{E}^t, \mathbf{N}$ denote the wall echo matrix, target echo matrix, and noise matrix. m, n denote sampling numbers of fast time t_r and slow time t_a. Hence, a low-rank and sparse optimization problem is established as (8) to estimate the wall clutter and target echoes

$$\min_{\mathbf{E}^w, \mathbf{E}^t}\left\{\mathrm{rank}\left(\mathbf{E}^w\right) + \lambda\left\|\mathbf{E}^t\right\|_0\right\} \tag{8}$$

where $\mathrm{rank}(\cdot)$ denotes the matrix rank, $\|\cdot\|_0$ denotes the ℓ_0-norm, and λ is a nonnegative parameter controlling the proportion of the low-rank constraint and the sparse constraint.

Since the optimization in (8) is an NP-hard problem, the nuclear norm $\|\mathbf{E}^w\|_*$ and ℓ_1-norm $\|\mathbf{E}^t\|_1$ are utilized to replace the matrix rank $\mathrm{rank}(\mathbf{E}^w)$ and ℓ_0-norm $\|\mathbf{E}^t\|_0$, respectively. Hence, the optimization problem is updated as

$$\min_{\mathbf{E}^w, \mathbf{E}^t}\left\{\|\mathbf{E}^w\|_* + \lambda\left\|\mathbf{E}^t\right\|_1\right\}. \tag{9}$$

III. Proposed JD-LRSD Method

The flowchart of the proposed JD-LRSD method is drawn in Fig. 4. The method designs a joint dual-channel low-rank and sparse model with the constraints of dual-channel similarity and wall clutter distribution. The dual-channel similarity constraint highlights the relationship between dual-channel echoes, while the wall clutter distribution constraint integrates the wall clutter distribution into target echo construction. The joint dual-channel low-rank and sparse optimization problem is further divided into several subproblems with the derived closed-form solutions. Then, the ADMM algorithm is adopted as the optimization solver for efficient wall clutter suppression in dual-channel UAV-mounted TTW-SAR.

A. Wall Clutter Analysis

Based on the homogeneous wall assumption, the wall clutter is mainly composed of electromagnetic waves reflected perpendicularly from the wall, including the front wall reflection, back wall reflection, and multiple wall reflections. The propagation distance is merely related to the wall parameters and the range interval between the EPCs and wall surfaces. Taking the front wall reflection as an example, the slant range from the ith EPC to the front wall surface y_{wall} is denoted as

$$R_i(t_a) = y_{\mathrm{wall}} - \hat{Y}_i(t_a) \tag{10}$$

where $\hat{Y}_i(t_a) = Y(t_a) + (3/2 - i)l\sin\theta_{\mathrm{y}}(t_a)\cos\theta_{\mathrm{r}}(t_a) + \Delta y(t_a)$ is merely related to the translational error in range $\Delta y(t_a)$ and attitude errors in yaw angle $\theta_{\mathrm{y}}(t_a)$ and roll angle $\theta_{\mathrm{r}}(t_a)$.

Due to motion errors, the practical flight path deviates from the planned path, leading to the nonzero difference of $R_i(t_a)$ and disrupting the low-rank characteristic of the wall clutter. Since the target echoes after wall attenuation are

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much weaker than wall clutter, the wall clutter possesses the highest correlation along the slow time direction. Hence, cross correlation is conducted on received echoes across PRTs for waveform alignment. After performing waveform alignment, the difference of $R_i(t_a)$ is approximately zero and the low-rank characteristic of the wall clutter is recovered.

Meanwhile, from (3), the similarity of the wall clutter across channels depends on the interchannel difference of $R_i(t_a)$, which is merely related to attitude errors $\theta_y(t_a), \theta_r(t_a)$

$$\Delta R(t_a) = \hat{Y}_1(t_a) - \hat{Y}_2(t_a) = l \sin\theta_y(t_a) \cos\theta_r(t_a). \tag{11}$$

In a monostatic dual-channel TTW radar system, the channel interval between EPCs is relatively small (centimeter level). According to the experimental configuration in Section IV, the channel interval is 0.03 m and the signal bandwidth is 0.408 GHz. Taking the poor flight condition with maximum rotation angles of 10°, the interchannel slant range errors caused by attitude errors are limited within 0.005 m, which are far below the range resolution. Hence, the attitude errors are negligible in the monostatic TTW radar system.

B. Joint Dual-Channel Low-Rank and Sparse Model

After performing waveform alignment on dual-channel UAV-mounted TTW-SAR, the received echoes of the ith channel based on the matrix notation are expressed as

$$\mathbf{E}_i = \mathbf{E}_i^w + \mathbf{E}_i^t + \mathbf{N}_i. \tag{12}$$

Since $\Delta R(t_a)$ of (3) is far below the range resolution and the attitude errors are negligible in the monostatic TTW radar system, the amplitude information of the wall clutter between channels is approximately the same. A dual-channel similarity constraint on the wall echo matrix is established as

$$\mathbf{E}_1^w = \mathbf{E}_2^w. \tag{13}$$

In TTW-SAR scenarios, the wall has a spatial extended occupancy along the azimuth direction, forming an image cluster with nonzero contiguous entries in high-resolution range profiles (HRRPs). Thus, performing incoherent superposition across PRTs, the superposition result is utilized to describe the wall clutter distribution and its mainlobe primarily covers the front wall reflection and the back wall reflection. Then, the mainlobe width is controlled by an exponential coefficient $p \in (0, 1]$ to cover multiple wall reflections. The window function is further conducted to remove the sidelobe energy from targets behind the wall. Hence, the wall clutter distribution weight of the ith channel is designed as

$$\mathbf{W}_i = \mathrm{norm}\left(\sum \mathbf{E}_i\right)^p \cdot \mathbf{W} \tag{14}$$

where $\sum$ is the incoherent superposition across PRTs, norm$(\cdot)$ is the matrix normalization scaling each element to the interval $[0, 1]$, and $\mathbf{W}$ is the window function.

Hence, the optimization problem for UAV-mounted TTW-SAR is established as (15), shown at the bottom of the page, where $\|\cdot\|_F$ is the Frobenius norm of the matrix and $\odot$ is the Hadamard product operator. $\lambda_1, \lambda_2, \eta_1, \eta_2, \eta_3, \eta_4$, and η_5 are adjustable nonnegative parameters controlling the proportion of each component, where λ_1 and λ_2 belong to the sparse component, η_1 and η_2 belong to the residual component, η_3 belongs to the dual-channel similarity constraint, and η_4 and η_5 belong to the wall clutter distribution constraint. Due to dual-channel similarity, wall clutter and target echoes are similar in amplitude across channels. Thus, hyperparameters are set as $\lambda_1 = \lambda_2$, $\eta_1 = \eta_2$, and $\eta_4 = \eta_5$.

C. ADMM-Based Wall Clutter Suppression Method

The ADMM algorithm is adopted as the optimization solver, which divides the optimization problem of (15) into a sequence of subproblems and performs the alternating update in a recursive manner. The detailed solution process of the ADMM framework is provided in the Appendix.

During the kth iteration, the wall echo matrices and target echo matrices are updated as (16a) and (16b) and (17a) and (17b), as shown at the bottom of the next page, which are denoted as

$$\mathbf{E}_1^{w,k+1} = \mathbf{U}_{\mathbf{Y}_1} \frac{\max\left(\mathbf{\Sigma}_{\mathbf{Y}_1} - \mathbf{I}, 0\right)}{\eta_1 + \eta_3} \mathbf{V}_{\mathbf{Y}_1}^{\mathrm{T}} \tag{16a}$$

$$\mathbf{E}_2^{w,k+1} = \mathbf{U}_{\mathbf{Y}_2} \frac{\max\left(\mathbf{\Sigma}_{\mathbf{Y}_2} - \mathbf{I}, 0\right)}{\eta_2 + \eta_3} \mathbf{V}_{\mathbf{Y}_2}^{\mathrm{T}} \tag{16b}$$

with the following notation definitions:

$$\mathbf{Y}_1 = \eta_1 \mathbf{E}_1 - \eta_1 \mathbf{E}_1^{t,k} + \eta_3 \mathbf{E}_2^{w,k} + \mathbf{Z}_1^k - \mathbf{Z}_3^k \tag{18a}$$

$$\mathbf{Y}_2 = \eta_2 \mathbf{E}_2 - \eta_2 \mathbf{E}_2^{t,k} + \eta_3 \mathbf{E}_1^{w,k+1} + \mathbf{Z}_2^k + \mathbf{Z}_3^k \tag{18b}$$

where $\mathbf{I}$ is the identity matrix and $\mathbf{\Sigma}_{\mathbf{Y}_i}$ is diagonal matrices after the eigenvalue decomposition $\mathbf{Y}_i = \mathbf{U}_{\mathbf{Y}_i} \mathbf{\Sigma}_{\mathbf{Y}_i} \mathbf{V}_{\mathbf{Y}_i}^{\mathrm{T}}$.

Then, the alternating updating processes of the Lagrange multipliers $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3, \mathbf{Z}_4, \mathbf{Z}_5 \in \mathbb{R}^{m \times n}$ are given by

$$\mathbf{Z}_1^{k+1} = \mathbf{Z}_1^k + \eta_1 \left(\mathbf{E}_1 - \mathbf{E}_1^{w,k+1} - \mathbf{E}_1^{t,k+1}\right) \tag{19a}$$

$$\mathbf{Z}_2^{k+1} = \mathbf{Z}_2^k + \eta_2 \left(\mathbf{E}_2 - \mathbf{E}_2^{w,k+1} - \mathbf{E}_2^{t,k+1}\right) \tag{19b}$$

$$\mathbf{Z}_3^{k+1} = \mathbf{Z}_3^k + \eta_3 \left(\mathbf{E}_1^{w,k+1} - \mathbf{E}_2^{w,k+1}\right) \tag{19c}$$

$$\mathbf{Z}_4^{k+1} = \mathbf{Z}_4^k + \eta_4 \left(\mathbf{W}_1 \odot \mathbf{E}_1^{t,k+1}\right) \tag{19d}$$

$$\mathbf{Z}_5^{k+1} = \mathbf{Z}_5^k + \eta_5 \left(\mathbf{W}_2 \odot \mathbf{E}_2^{t,k+1}\right). \tag{19e}$$

After the alternating update, the termination condition of the ADMM algorithm is set as

$$\mathrm{T}_1 : \max\left(\left.\frac{\left\|\mathbf{E}_\mathrm{i} - \mathbf{E}_\mathrm{i}^\mathrm{w} - \mathbf{E}_\mathrm{i}^\mathrm{t}\right\|_\mathrm{F}}{\|\mathbf{E}_\mathrm{i}\|_\mathrm{F}}\right|_{\mathrm{i}=1,2}\right) \le \varepsilon_1 \tag{20a}$$

$$\min_{\mathbf{E}_1^w, \mathbf{E}_2^w, \mathbf{E}_1^t, \mathbf{E}_2^t} \left\{ \begin{array}{l} \left\|\mathbf{E}_1^w\right\|_* + \left\|\mathbf{E}_2^w\right\|_* + \lambda_1 \left\|\mathbf{E}_1^t\right\|_1 + \lambda_2 \left\|\mathbf{E}_2^t\right\|_1 \\ + \dfrac{\eta_1}{2} \left\|\mathbf{E}_1 - \mathbf{E}_1^w - \mathbf{E}_1^t\right\|_F^2 + \dfrac{\eta_2}{2} \left\|\mathbf{E}_2 - \mathbf{E}_2^w - \mathbf{E}_2^t\right\|_F^2 \\ + \dfrac{\eta_3}{2} \left\|\mathbf{E}_1^w - \mathbf{E}_2^w\right\|_F^2 + \dfrac{\eta_4}{2} \left\|\mathbf{W}_1 \odot \mathbf{E}_1^t\right\|_F^2 + \dfrac{\eta_5}{2} \left\|\mathbf{W}_2 \odot \mathbf{E}_2^t\right\|_F^2 \end{array} \right\}. \tag{15}$$

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$$\mathrm{T}_2 : \max\left(\left.\frac{\left\|\mathbf{E}_{\mathrm{i}}^{\mathrm{t}+1}-\mathbf{E}_{\mathrm{i}}^{\mathrm{t}}\right\|_{\mathrm{F}}}{\|\mathbf{E}_{\mathrm{i}}\|_{\mathrm{F}}}\right|_{\mathrm{i}=1,2}\right) \le \varepsilon_2 \tag{20b}$$

where ε_1 and ε_2 are thresholds for termination.

The termination condition T_1 constrains the normalized residual energy belonging to the noise component and modeling errors. Its convergence indicates the noise matrix realizes an extremely low level. The termination condition T_2 constrains the target energy variation between adjacent iterations. Its convergence indicates the algorithm realizes iterative stability and further iterations are unnecessary. Due to the possible misassignment of the noise component, the convergence is not equivalent to the optimal SINR in wall clutter suppression. The thresholds $\varepsilon_1, \varepsilon_2$ are both set beyond theoretical convergence bounds to balance stability and efficiency.

The time complexity is further analyzed as follows. Based on dual-channel signal matrices $\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^{m\times n}$, the dominant computational burden of each iteration comes from the eigenvalue decomposition on $\mathbf{Y}_1$ and $\mathbf{Y}_2$ in (16a) and (16b) with the time complexity of $O(mn \cdot \min(m,n))$. Element-wise soft-thresholding operations in (17a) and (17b), and Lagrange multiplier updates in (19a)–(19e) only introduce negligible complexity of $O(mn)$. Hence, assuming that the proposed method converges at the Kth iteration, the time complexity of JD-LRSD is formulated as $O(2Kmn \cdot \min(m,n))$.

Through the iterative update in the ADMM-based wall clutter suppression, target echo matrices $\mathbf{E}_1^t, \mathbf{E}_2^t$ are estimated and the HRRPs are recovered by reverse waveform alignment. To evaluate the wall clutter suppression performance, the back projection (BP) algorithm is adopted for SAR imaging and the SINR criterion is selected for performance evaluation. Since the refraction effect in the TTW scenario would cause shifts and blurs for targets behind the wall in (5), the priori known wall parameters are introduced to compensate the refraction effect and guarantee the effectiveness of the evaluation by SINR [51], [53], [54]. The pseudocode description of the proposed JD-LRSD method is summarized in Algorithm 1.

IV. Simulations and Experiments

To verify the wall clutter suppression performance of the proposed method, numerical simulations and field experiments are provided in this section.

A. Numerical Simulation

Adopting the simulation parameters provided in Table I, numerical simulations are conducted based on the gprMax software [55]. The adopted waveform is LFMCW and the signal bandwidth is 0.5 GHz with the start and stop frequencies of 2 and 2.5 GHz. In the simulated monostatic dual-channel radar system, the transmitting antenna and the receiving antenna are assumed to be co-located and the dual-channel EPCs are parallel to the azimuth direction with the channel interval of 0.03 m. The geometric configuration is drawn in Fig. 5, where the presupposed trajectory is from (−4, 0, 0) to (4, 0, 0) m with the PRT number of 128. Wall parameters are denoted as the thickness of d and the relative permittivity of ε_r. The wall is parallel to the azimuth direction and the front wall surface is located at the range position of 5 m. The simulation sets four targets with the same scattering coefficient behind the wall. Assuming the spacing between wall and targets is r, the target positions are $(\pm 1.5,\ 5+d+r)$ m and $(\pm 1.5,\ 7+d+r)$ m. The wall clutter of (10) is merely related to the translational error in range and attitude errors in yaw angle and roll angle. Since the attitude errors are negligible in the monostatic TTW radar system, the translational error in range $\Delta y(t_a)$ is only adopted in simulations. A sinusoidal variation with the range of $[-0.03, 0.03]$ m is further employed as the kinematic model for the UAV's center of mass.

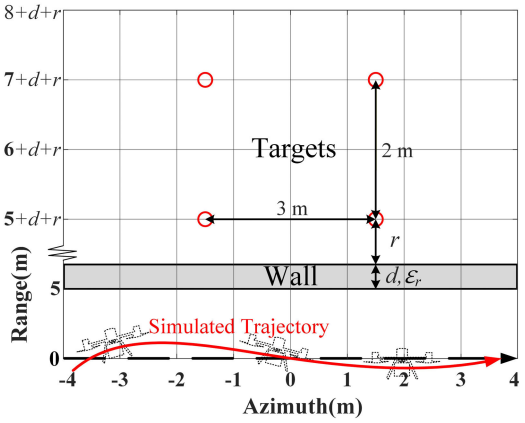


Fig. 5. Simulated geometric configuration.

TABLE I
Simulation Parameters

Parameter	Value
Waveform	LFMCW
Start / Stop Frequency	2 GHz / 2.5 GHz
Signal Bandwidth	0.5 GHz
PRT Number	128
Presupposed Trajectory	(-4,0,0) m→(4,0,0) m
Motion Model	Sinusoidal Curve
Target Number	4

$$\mathbf{E}_1^{t,k+1} = \operatorname{sign}\left(\frac{\eta_1\mathbf{E}_1 - \eta_1\mathbf{E}_1^{w,k+1} + \mathbf{Z}_1^k - \mathbf{W}_1\odot\mathbf{Z}_4^k}{\eta_1+\eta_4\mathbf{W}_1\odot\mathbf{W}_1}\right)\max\left(\left|\frac{\eta_1\mathbf{E}_1 - \eta_1\mathbf{E}_1^{w,k+1} + \mathbf{Z}_1^k - \mathbf{W}_1\odot\mathbf{Z}_4^k}{\eta_1+\eta_4\mathbf{W}_1\odot\mathbf{W}_1}\right| - \frac{\lambda_1}{\eta_1+\eta_4\mathbf{W}_1\odot\mathbf{W}_1}, 0\right) \tag{17a}$$

$$\mathbf{E}_2^{t,k+1} = \operatorname{sign}\left(\frac{\eta_2\mathbf{E}_2 - \eta_2\mathbf{E}_2^{w,k+1} + \mathbf{Z}_2^k - \mathbf{W}_2\odot\mathbf{Z}_5^k}{\eta_2+\eta_5\mathbf{W}_2\odot\mathbf{W}_2}\right)\max\left(\left|\frac{\eta_2\mathbf{E}_2 - \eta_2\mathbf{E}_2^{w,k+1} + \mathbf{Z}_2^k - \mathbf{W}_2\odot\mathbf{Z}_5^k}{\eta_2+\eta_5\mathbf{W}_2\odot\mathbf{W}_2}\right| - \frac{\lambda_2}{\eta_2+\eta_5\mathbf{W}_2\odot\mathbf{W}_2}, 0\right) \tag{17b}$$

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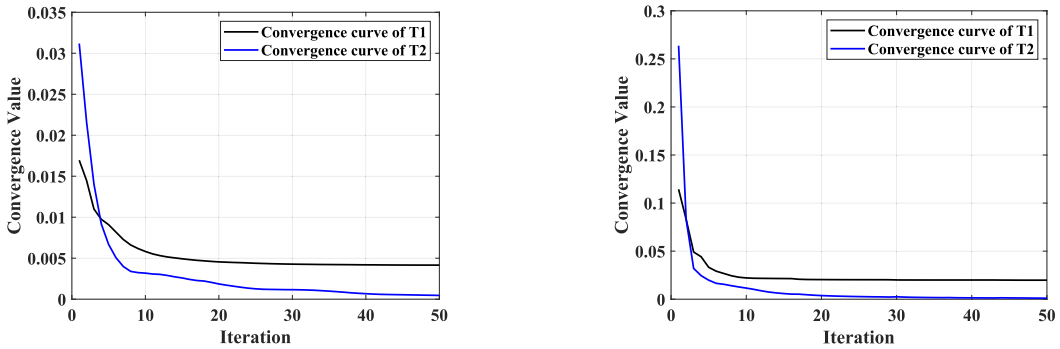


Fig. 6. Convergence procedure of the ADMM algorithm. (a) Simulation. (b) Experiment.

Algorithm 1 Proposed JD-LRSD Method

Input: Maximum iteration number K;
Dual-channel radar echoes $\mathbf{E}_1, \mathbf{E}_2$;
Convergence thresholds $\varepsilon_1, \varepsilon_2$.
Output: Wall echo matrices $\mathbf{E}_1^w, \mathbf{E}_2^w$;
Target echo matrices $\mathbf{E}_1^t, \mathbf{E}_2^t$.

Initialize: Lagrange multipliers $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3, \mathbf{Z}_4, \mathbf{Z}_5$;
Adjustable hyperparameters $\lambda_1, \lambda_2, \eta_1, \eta_2, \eta_3, \eta_4, \eta_5$.

Waveform alignment and amplitude extraction;
Construct wall clutter distribution weights $\mathbf{W}_1, \mathbf{W}_2$;
for $k \leftarrow 1$ **to** K **do**
  Eigenvalue decomposition on $\mathbf{Y}_1$ in (18a);
  Update $\mathbf{E}_1^{w,k+1}$ by (16a);
  Eigenvalue decomposition on $\mathbf{Y}_2$ in (18b);
  Update $\mathbf{E}_2^{w,k+1}$ by (16b);
  Update $\mathbf{E}_1^{t,k+1}$ by (17a);
  Update $\mathbf{E}_2^{t,k+1}$ by (17b);
  Update Lagrange multipliers by (19a)-(19e);
  if *termination condition is satisfied* **then**
    Break;
  end
end
Phase modulation and reverse waveform alignment.

In the regularization parameter selection, due to dual-channel similarity, there is an assumption $\lambda_1 = \lambda_2, \eta_1 = \eta_2$, and $\eta_4 = \eta_5$. When setting $\eta_3 = \eta_4 = \eta_5 = 0$, the optimization problem in (15) is transformed into an conventional three-term stable decomposition. Thus, constraints of sparse components and residual components λ_1, η_1 are first adjusted, while constraints of dual-channel similarity and wall clutter distribution η_3, η_4 are further adjusted. Simulated hyperparameters are set as $\lambda_1 = \lambda_2 = 1e^{-3}, \eta_1 = \eta_2 = 0.5, \eta_3 = 1$, and $\eta_4 = \eta_5 = 10$. Taking $d = 0.2$ m, $\varepsilon_r = 6$, and $r = 2.8$ m as an example, the convergence procedure of the ADMM algorithm is shown in Fig. 6(a), where T_1 and T_2 converge at the 20th and 40th iterations, respectively. When T_1 meets the termination condition, T_2 still remains unconverged and subsequent iterations would misclassify target signals as low-rank wall clutter. Meanwhile, the lowest noise level of T_1 is not equivalent to the optimal performance in the final image. Thus, we set the termination limited by T_1 with the threshold of $\varepsilon_1 = 8e^{-3}$.

Considering the wall clutter suppression performance is sensitive to wall characteristics and the spacing between wall and targets, numerical simulations are splitted into five parts. The first three parts are to separately discuss the effect of wall thickness, permittivity, and heterogeneity on performance. The third part centers on the discussion of spacing between wall and targets. At last, contrast tests are performed to verify the accuracy and effectiveness of the proposed JD-LRSD method.

1) Effect of Wall Thickness: The simulated geometric model adopting $d \in [0.1, 0.4]$ m is set to explore the effect of wall thickness on wall clutter suppression performance. The relative permittivity ε_r is set as 6 and the simulated four targets are located at ($\pm$1.5, 8) and ($\pm$1.5, 10) m. The simulation results are drawn in Fig. 7(a), where the average SINRs of the dual-channel SAR images after wall clutter suppression are within [21.5, 23] dB. The inversion results indicate that the proposed method is robust for different wall thicknesses.

2) Effect of Wall Permittivity: The simulated geometric model adopting $\varepsilon_r \in [2, 16]$ is set to explore the effect of wall permittivity on wall clutter suppression performance. The thickness d is set as 0.2 m and the simulated four targets are located at ($\pm$1.5, 8) and ($\pm$1.5, 10) m. As shown in Fig. 7(b), the average SINRs of the simulation results are within [20, 24] dB, indicating that the proposed method is robust for different wall permittivity.

3) Effect of Wall Heterogeneity: Adopting the wall thickness $d = 0.2$ m and the spacing $r = 2.8$ m, the simulated geometric model with the heterogeneous wall is constructed to explore the effect of wall heterogeneity on wall clutter suppression performance. The heterogeneous wall is composed of voxels with relative permittivity $\varepsilon_r \in [4, 8]$.

The simulation results are provided in Fig. 8. Performing the proposed method, the low-rank component of Fig. 8(a) denotes the wall clutter and the sparse component of Fig. 8(b) denotes the target echoes. Wall clutter mainly distributed within [4, 6] m possesses the maximum amplitude, while target echoes are located at nearly [6, 10] m in the slant range direction, suffering from low SINR. The original SAR images of the dual-channel data are shown in Fig. 8(f) and (g), where the four targets are defocused due to the wall

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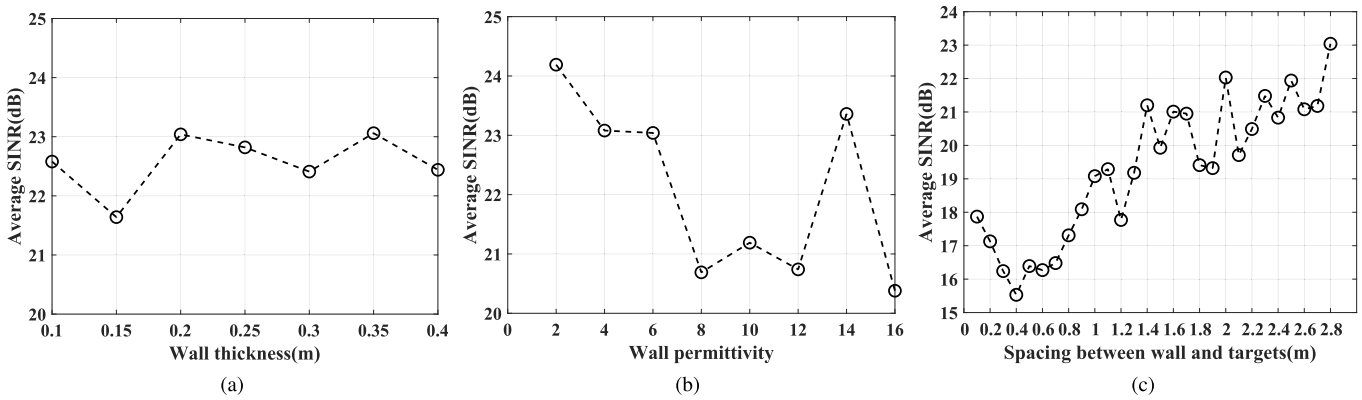


Fig. 7. Performance in wall clutter suppression of the proposed method. (a) Effect of wall thickness. (b) Effect of wall permittivity. (c) Effect of spacing between wall and targets.

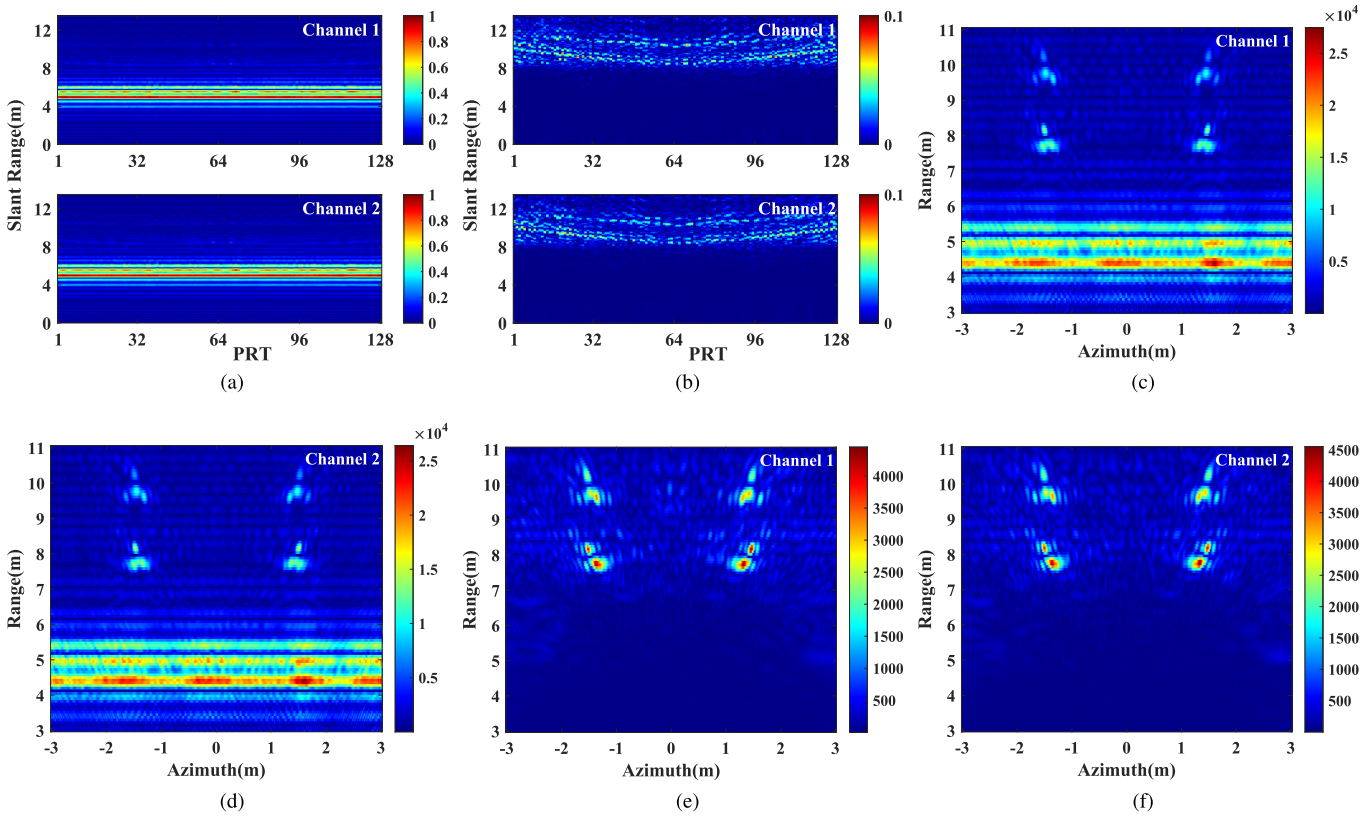


Fig. 8. Simulation results of the proposed method in the case of the heterogeneous wall. (a) Estimated low-rank component. (b) Estimated sparse component. (c) and (d) Original SAR images of channels 1 and 2. (e) and (f) SAR images of channels 1 and 2 after wall clutter suppression.

heterogeneity. As shown in Fig. 8(c) and (d), the proposed method achieves effective wall clutter suppression. Wall clutter is effectively eliminated and the dual-channel SINRs are 17.17 and 17.99 dB.

4) Effect of Spacing Between Wall and Targets: The simulated geometric model adopting $r \in [0.1, 2.8]$ m is set to explore the effect of spacing between wall and targets on wall clutter suppression performance. Assuming the wall parameters of $d = 0.2$ m and $\varepsilon_r = 6$, the simulation results are drawn in Fig. 7(c). As the target gets relatively closer to the wall, target echoes are overwhelmed by the wall clutter, causing deteriorated wall clutter suppression performance and declined average SINRs.

The simulation results of the spacing $r = 0.1$ m are provided in Fig. 9, where Fig. 9(a) and (b) is estimated low-rank and sparse components. From the original SAR images in Fig. 9(c) and (d), the two targets located at $[\pm 1.5, 5.3]$ are overwhelmed by the wall clutter. After wall clutter suppression, the dual-channel SINRs are 16.80 and 18.94 dB in Fig. 9(e) and (f). It demonstrates that the proposed method is robust for the small spacing cases.

5) Ablation and Contrast Tests: Taking $d = 0.2$ m, $\varepsilon_r = 6$, and $r = 2.8$ m as an example, the geometric model with the homogeneous wall is adopted. Ablation tests are first conducted to verify the contributions of the individual constraints in wall clutter suppression. Then, contrast tests

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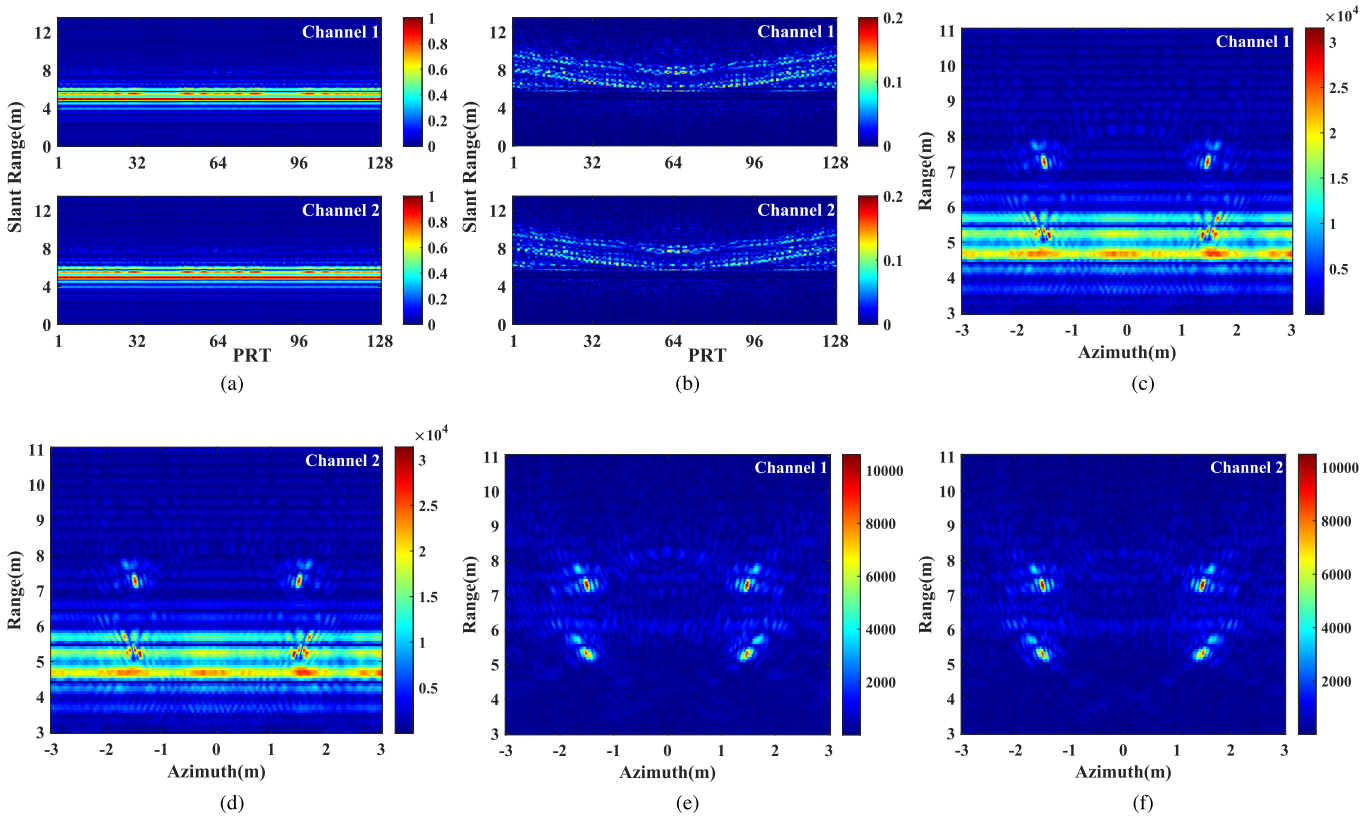


Fig. 9. Simulation results of the proposed method in the case of the spacing of 0.1 m. (a) Estimated low-rank component. (b) Estimated sparse component. (c) and (d) Original SAR images of channels 1 and 2. (e) and (f) SAR images of channels 1 and 2 after wall clutter suppression.

TABLE II
PERFORMANCE IN WALL CLUTTER SUPPRESSION

SINR(dB)	JD-LRSD	Ablation Tests		Contrast Tests	
		η_3=0	η_4=η_5=0	RPCA [16]	WNNM-TVR [20], [22]
Channel 1 in Simulations	**23.19**	21.73	19.70	18.25	18.38
Channel 2 in Simulations	**22.90**	22.27	20.25	18.68	18.18
Channel 1 in Experiments	**17.05**	15.78	10.10	8.42	9.15
Channel 2 in Experiments	**14.91**	12.47	6.87	4.98	3.38

are further conducted. RPCA [16] and WNNM with total variation regularization [20], [22] are selected as comparisons, which are performed independently on each channel. Since the contrast methods do not contain any processing for motion errors, the dual-channel data after waveform alignment are taken as the input data.

For ablation tests, the individual wall clutter distribution constraint is obtained by setting $\eta_3 = 0$, while the individual dual-channel similarity constraint is obtained by setting $\eta_4 = \eta_5 = 0$. Since the slant range discrepancy between dual-channel wall clutters is far below the range resolution, wall echo matrices $\mathbf{E}_1^w, \mathbf{E}_2^w$ inherently exhibit high similarity even without the dual-channel similarity constraint. As quantified in Table II, the wall clutter distribution constraint contributes the primary performance improvement for the proposed method.

For contrast tests, RPCA and WNNM-TVR decompose the data matrix into low-rank wall clutters [Figs. 10(a) and 11(a)] and sparse target echoes [Figs. 10(b) and 11(b)]. Since simulated data are similar across channels and contrast methods operate independently on each channel, only SAR images of channel 1 are given in Figs. 10(c) and 11(c). As shown in Table II, contrast methods achieve similar performance with average SINRs of 18.47 and 18.28 dB, while the proposed JD-LRSD realizes the best performance.

In terms of computational efficiency, the time complexity of JD-LRSD, RPCA, and WNNM-TVR for dual-channel data are $O(2Kmn \cdot \min(m,n))$, $O(2Kmn \cdot \min(m,n))$, and $O(4Kmn \cdot \min(m,n) + 2k_0\ mn)$, where k_0 is the iteration number of fast projected gradient algorithm. However, the actual running time is also related to the iteration number when the algorithm converges. The run time results of JD-LRSD, RPCA, and WNNM-TVR are 1.87 s with six iterations, 3.64 s

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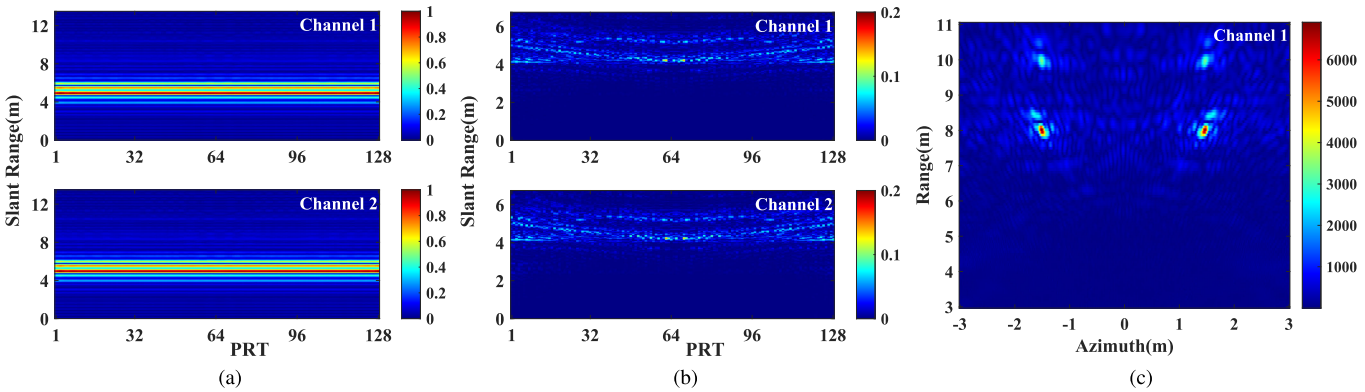


Fig. 10. Simulation results of RPCA. (a) Estimated low-rank component. (b) Estimated sparse component. (c) SAR image of channel 1 after clutter suppression.

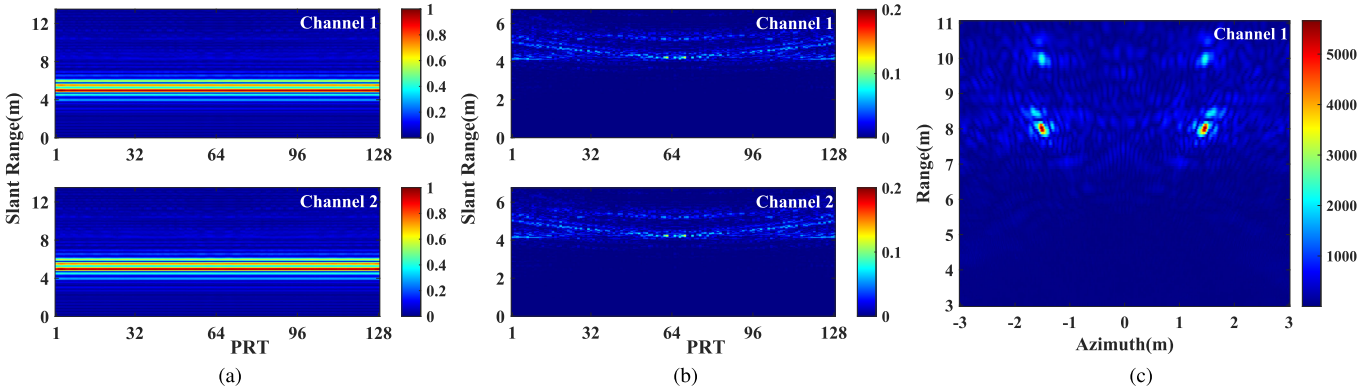


Fig. 11. Simulation results of WNNM-TVR. (a) Estimated low-rank component. (b) Estimated sparse component. (c) SAR image of channel 1 after clutter suppression.

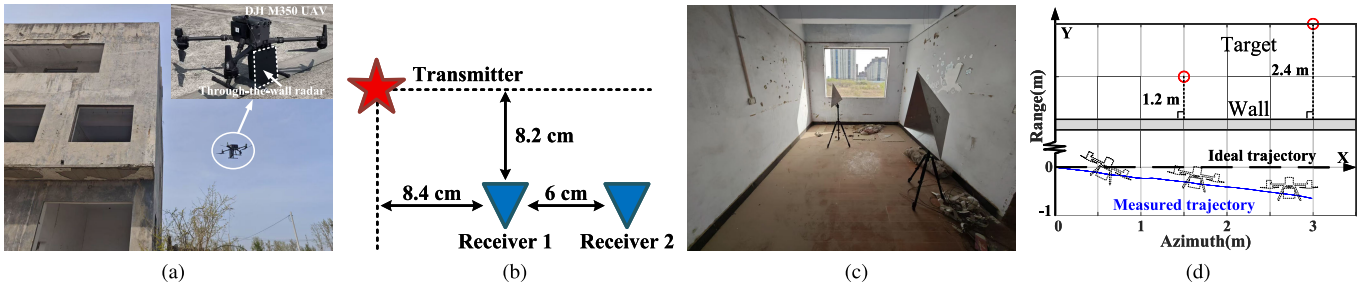


Fig. 12. Field experiments. (a) UAV-mounted TTW-SAR scenario. (b) Antenna configuration. (c) Targets behind the wall. (d) Geometric configuration.

with ten iterations, and 6.01 s with five iterations, verifying the practical applicability of the proposed method.

B. Field Experiment

Adopting experimental parameters provided in Table III, field experiments are conducted based on a DJI M350 quadrotor in Fig. 12(a). The quadrotor is equipped with INS/GPS for real-time location information acquisition and the monostatic dual-channel TTW radar system is installed beneath the quadrotor. The radar system adopts LFMCW as the waveform and the signal bandwidth is 0.408 GHz with the start and stop frequencies of 2.67 and 3.078 GHz. The antenna configuration is shown in Fig. 12(b) and the dual-channel EPCs are parallel to the azimuth direction.

TABLE III
EXPERIMENTAL PARAMETERS

Parameter	Value
Waveform	LFMCW
Signal Bandwidth	0.408 GHz
Start / Stop Frequency	2.67 GHz / 3.078 GHz
Antenna Configuration	One-Transmit-Two-Receive
Wall Category	Brick wall
Wall Thickness	0.24 m
Target Number	2

As shown in Fig. 12(a), the UAV flies around the second floor of the building, maintaining a close distance to the wall

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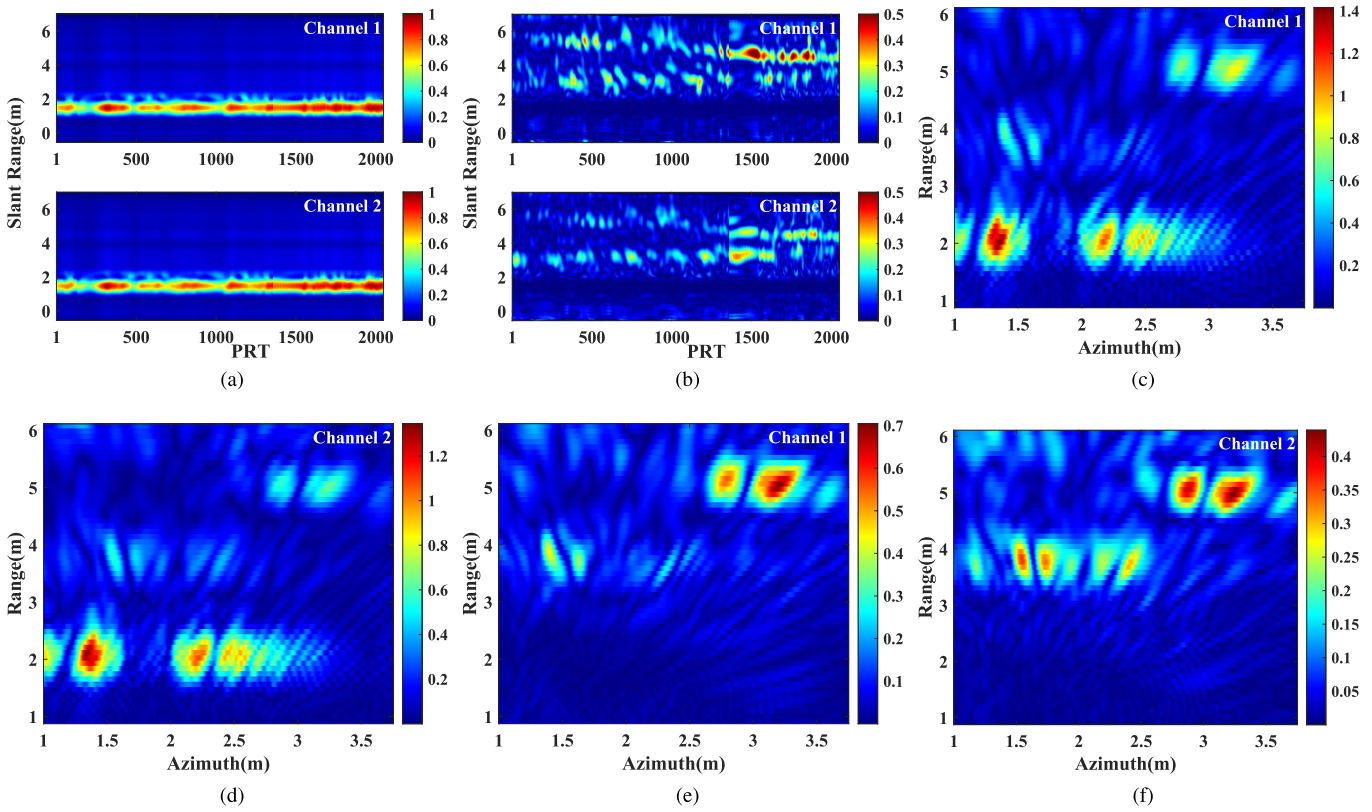


Fig. 13. Experimental results of the proposed method. (a) Estimated low-rank component. (b) Estimated sparse component. (c) and (d) Original SAR images of channels 1 and 2. (e) and (f) SAR images of channels 1 and 2 after wall clutter suppression.

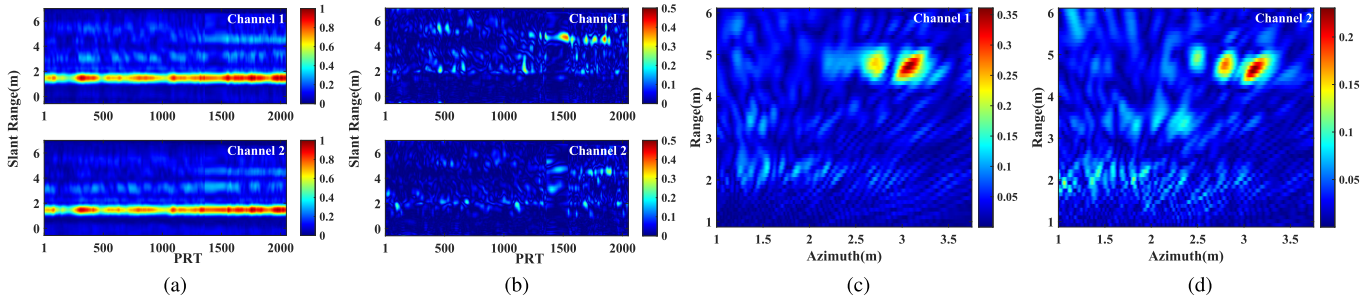


Fig. 14. Experimental results of RPCA. (a) Estimated low-rank component. (b) Estimated sparse component. (c) and (d) SAR images of channels 1 and 2 after wall clutter suppression.

surface. The wall is constructed of brick with the thickness of 0.24 m, oriented along the azimuth direction. In Fig. 12(c), there are two corner reflectors set behind the wall and the two reflectors are 1.2 and 2.4 m away from the back wall surface with an azimuth interval of 1.5 m. The ideal trajectory is parallel to the azimuth direction, while the INS/GPS-measured trajectory progressively deviates from the ideal trajectory shown in Fig. 12(d).

In the regularization parameter selection, experimental hyperparameters are set as $\lambda_1 = \lambda_2 = 5e^{-4}, \eta_1 = \eta_2 = 0.01, \eta_3 = 0.5$, and $\eta_4 = \eta_5 = 5$. In the experimental convergence procedure of Fig. 6(b), T_1 and T_2 converge at the 10th and 40th iterations, and the termination threshold is set as $\varepsilon_1 = 2.5e^{-2}$.

Experimental results of the proposed method are provided in Fig. 13. The low-rank component in Fig. 13(a) denotes the wall clutter and the sparse component in Fig. 13(b) denotes the target echoes. Due to the waveform alignment, the low-rank characteristic of the wall clutter is well-recovered. Performing the BP algorithm, the original SAR images in Fig. 13(c) and (d) suffer from low SINR. The proposed method effectively suppresses the wall clutter with imaging results of Fig. 13(e) and (f), where the wall clutter located at 2 m is significantly eliminated. As provided in Table II, the dual-channel SINRs after wall clutter suppression are 17.05 and 14.91 dB.

Then, ablation and contrast tests are further conducted. Similar to numerical simulations, experimental ablation results in Table II demonstrate the effectiveness of each individual constraint, where the wall clutter distribution constraint contributes the primary performance improvement. In contrast tests, RPCA and WNNM-TVR are adopted as comparisons

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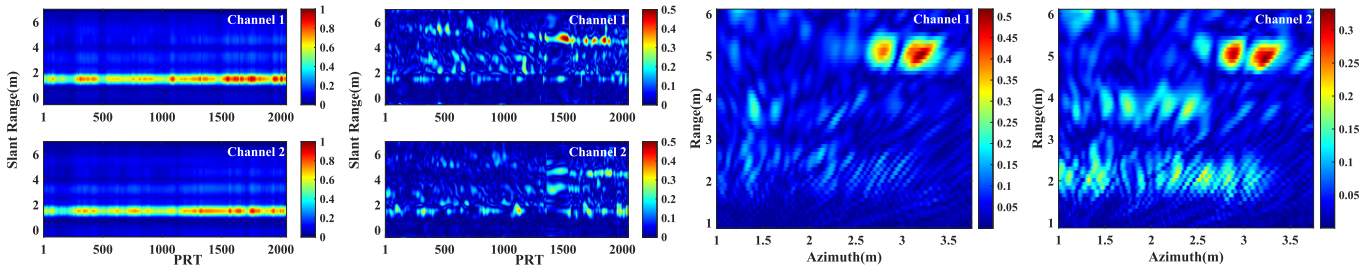


Fig. 15. Experimental results of WNNM-TVR. (a) Estimated low-rank component. (b) Estimated sparse component. (c) and (d) SAR images of channels 1 and 2 after wall clutter suppression.

with processing results shown in Figs. 14 and 15. The decomposed low-rank wall clutter is presented in Figs. 14(a) and 15(a), and the decomposed sparse target echoes are presented in Figs. 14(b) and 15(b). The decomposition indicates that RPCA erroneously allocates a portion of target echoes into the low-rank component, while WNNM-TVR identifies more wall clutters as the sparse component. The final dual-channel SAR images are drawn in Figs. 14(c) and (d) and 15(c) and (d), demonstrating contrast methods have limited wall clutter suppression performance. As shown in Table II, the average SINRs processed by RPCA and WNNM-TVR are 6.70 and 6.27 dB with the running times of 38.45 and 87.70 s in five iterations. Consequently, the proposed JD-LRSD method demonstrates its superior performance in wall clutter suppression for UAV-mounted TTW-SAR, outperforming the contrast methods by approximately 9 dB with the running times of 72.58 s in nine iterations.

V. Conclusion

UAV-mounted TTW-SAR is a critical technology for urban structure penetration and wall clutter suppression is a key prerequisite for its applications. In this article, a JD-LRSD method for wall clutter suppression based on UAV-mounted TTW-SAR has been proposed. In the proposed method, the joint dual-channel low-rank and sparse model combined with the dual-channel similarity constraint and the wall clutter distribution constraint has been developed as the optimization problem. Then, the ADMM-based wall clutter suppression method has been designed, updating each component in a recursive manner. Closed-form solutions of each subproblem have been derived as explicit expressions. Finally, simulations and experiments have demonstrated the effectiveness of the proposed JD-LRSD method for UAV-mounted TTW-SAR, making it a promising solution for wall clutter suppression in multichannel radar systems.

During the exploration of wall clutter suppression in UAV-mounted TTW-SAR, several potential improvements about the future research direction are worth considering. First, the joint low-rank and sparse strategy is established on a monostatic dual-channel radar system, which requires extension to the multichannel (more than two channels) and multimonostatic configuration. Second, if the channel interval is large enough that platform attitude errors are nonnegligible, motion error estimation and compensation are required to be considered. Although the proposed method has achieved satisfactory results, future research would focus on more robust and generalized wall clutter suppression for practical applications. On the other hand, as a preprocessing step in UAV-mounted TTW-SAR, the proposed wall clutter suppression technique would become the impetus for the development of other tasks, such as target detection, location, and tracking.

Appendix
Solution Process of the ADMM Framework

Adopting the ADMM algorithm for solving the optimization problem in (15), the augmented Lagrangian function is constructed as (21), shown at the bottom of the next page, where $\mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3, \mathbf{Z}_4, \mathbf{Z}_5 \in \mathbb{R}^{m\times n}$ are the Lagrange multipliers.

Then, the optimization problem is divided into a sequence of subproblems and the ADMM algorithm updates each component in a recursive manner by solving subproblems. During the kth iteration, the alternating updating processes of the optimization subproblems are given by

$$\mathbf{E}_1^{w,k+1} = \arg\min_{\mathbf{E}_1^w} L\left(\mathbf{E}_1^w, \mathbf{E}_2^{w,k}, \mathbf{E}_1^{t,k}, \mathbf{E}_2^{t,k}\right) \tag{22a}$$

$$\mathbf{E}_2^{w,k+1} = \arg\min_{\mathbf{E}_2^w} L\left(\mathbf{E}_1^{w,k+1}, \mathbf{E}_2^{w}, \mathbf{E}_1^{t,k}, \mathbf{E}_2^{t,k}\right) \tag{22b}$$

$$\mathbf{E}_1^{t,k+1} = \arg\min_{\mathbf{E}_1^t} L\left(\mathbf{E}_1^{w,k+1}, \mathbf{E}_2^{w,k+1}, \mathbf{E}_1^{t}, \mathbf{E}_2^{t,k}\right) \tag{22c}$$

$$\mathbf{E}_2^{t,k+1} = \arg\min_{\mathbf{E}_2^t} L\left(\mathbf{E}_1^{w,k+1}, \mathbf{E}_2^{w,k+1}, \mathbf{E}_1^{t,k+1}, \mathbf{E}_2^{t}\right). \tag{22d}$$

A. Closed-Form Solution of the Subproblem in (22a)

Substituting the augmented Lagrangian function into subproblem (22a), the wall echo matrix of channel 1 $\mathbf{E}_1^{w,k+1}$ is updated as (23a), shown at the bottom of the next page, and corresponding derivative is derived as

$$\begin{aligned}&\frac{\partial L\left(\mathbf{E}_1^w, \mathbf{E}_2^{w,k}, \mathbf{E}_1^{t,k}, \mathbf{E}_2^{t,k}\right)}{\partial \mathbf{E}_1^w}\\ &= \frac{\partial \left\|\mathbf{E}_1^w\right\|_*}{\partial \mathbf{E}_1^w} + (\eta_1+\eta_3)\,\mathbf{E}_1^w - \mathbf{Y}_1\\ &= \mathbf{U}_{\mathbf{Y}_1}\left((\eta_1+\eta_3)\,\mathbf{\Sigma}_{\mathbf{E}_1^w} - \mathbf{\Sigma}_{\mathbf{Y}_1} + \mathbf{I}\right)\mathbf{V}_{\mathbf{Y}_1}^{\mathrm{T}}\end{aligned} \tag{24}$$

where $\mathbf{Y}_1 = \eta_1\mathbf{E}_1 - \eta_1\mathbf{E}_1^{t,k} + \eta_3\mathbf{E}_2^{w,k} + \mathbf{Z}_1^k - \mathbf{Z}_3^k$. Based on the eigenvalue decomposition $\mathbf{E}_1^w = \mathbf{U}_{\mathbf{E}_1^w}\mathbf{\Sigma}_{\mathbf{E}_1^w}\mathbf{V}_{\mathbf{E}_1^w}^{\mathrm{T}}$ and $\mathbf{Y}_1 =$

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$\mathbf{U}_{\mathbf{Y}_1}\mathbf{\Sigma}_{\mathbf{Y}_1}\mathbf{V}_{\mathbf{Y}_1}^{\mathrm{T}}$, the optimal solution to the subproblem is obtained when $\mathbf{E}_1^w$ and $\mathbf{Y}_1$ share same singular vectors, namely $\mathbf{U}_{\mathbf{E}_1^w} = \mathbf{U}_{\mathbf{Y}_1}, \mathbf{V}_{\mathbf{E}_1^w} = \mathbf{V}_{\mathbf{Y}_1}$.

Since $\mathbf{\Sigma}_{\mathbf{E}_1^w}$ and $\mathbf{\Sigma}_{\mathbf{Y}_1}$ are both nonnegative diagonal matrices, setting the derivative to zero, the solution is derived as

$$\mathbf{E}_1^w = \mathbf{U}_{\mathbf{Y}_1}\mathbf{\Sigma}_{\mathbf{E}_1^w}\mathbf{V}_{\mathbf{Y}_1}^{\mathrm{T}} = \mathbf{U}_{\mathbf{Y}_1}\frac{\max\left(\mathbf{\Sigma}_{\mathbf{Y}_1} - \mathbf{I}, 0\right)}{\eta_1 + \eta_3}\mathbf{V}_{\mathbf{Y}_1}^{\mathrm{T}} \tag{25}$$

where $\mathbf{I}$ is the identity matrix.

B. Closed-Form Solution of the Subproblem in (22b)

Substituting the augmented Lagrangian function into the subproblem (22b), the wall echo matrix of channel 2 $\mathbf{E}_2^{w,k+1}$ is updated as (23b), shown at the bottom of the page, and the corresponding derivative is derived as

$$\begin{aligned}&\frac{\partial L\left(\mathbf{E}_1^{w,k+1}, \mathbf{E}_2^w, \mathbf{E}_1^{t,k}, \mathbf{E}_2^{t,k}\right)}{\partial \mathbf{E}_2^w}\\ &= \frac{\partial\left\|\mathbf{E}_2^w\right\|_*}{\partial \mathbf{E}_2^w} + (\eta_2 + \eta_3)\mathbf{E}_2^w - \mathbf{Y}_2\\ &= \mathbf{U}_{\mathbf{Y}_2}\left((\eta_2 + \eta_3)\mathbf{\Sigma}_{\mathbf{E}_2^w} - \mathbf{\Sigma}_{\mathbf{Y}_2} + \mathbf{I}\right)\mathbf{V}_{\mathbf{Y}_2}^{\mathrm{T}}\end{aligned} \tag{26}$$

where $\mathbf{Y}_2 = \eta_2\mathbf{E}_2 - \eta_2\mathbf{E}_2^{t,k} + \eta_3\mathbf{E}_1^{w,k+1} + \mathbf{Z}_2^k + \mathbf{Z}_3^k$. The eigenvalue decompositions are denoted as $\mathbf{E}_2^w = \mathbf{U}_{\mathbf{E}_2^w}\mathbf{\Sigma}_{\mathbf{E}_2^w}\mathbf{V}_{\mathbf{E}_2^w}^{\mathrm{T}}$ and $\mathbf{Y}_2 = \mathbf{U}_{\mathbf{Y}_2}\mathbf{\Sigma}_{\mathbf{Y}_2}\mathbf{V}_{\mathbf{Y}_2}^{\mathrm{T}}$.

Similar to (25), the solution is derived as

$$\mathbf{E}_2^w = \mathbf{U}_{\mathbf{Y}_2}\mathbf{\Sigma}_{\mathbf{E}_2^w}\mathbf{V}_{\mathbf{Y}_2}^{\mathrm{T}} = \mathbf{U}_{\mathbf{Y}_2}\frac{\max\left(\mathbf{\Sigma}_{\mathbf{Y}_2} - \mathbf{I}, 0\right)}{\eta_2 + \eta_3}\mathbf{V}_{\mathbf{Y}_2}^{\mathrm{T}}. \tag{27}$$

C. Closed-Form Solution of the Subproblem in (22c)

Substituting the augmented Lagrangian function into the subproblem (22c), the target echo matrix of channel 1 $\mathbf{E}_1^{t,k+1}$ is updated as (23c), shown at the bottom of the page. Performing variable substitution, the substitution matrices are denoted as $\mathbf{A}_1 = \mathbf{E}_1 - \mathbf{E}_1^{w,k+1} + \mathbf{Z}_1^k/\eta_1, \mathbf{A}_4 = \mathbf{Z}_4^k/\eta_4$.

Based on the independence of each matrix element in L, the optimization in (22c) is divided into the optimization of each matrix element, which is denoted as

$$\min_{e_{1,ij}^t}\left\{\lambda_1\left|e_{1,ij}^t\right| + \frac{\eta_1}{2}\left(a_{1,ij} - e_{1,ij}^t\right)^2 + \frac{\eta_4}{2}\left(w_{1,ij}e_{1,ij}^t + a_{4,ij}\right)^2\right\} \tag{28}$$

where $e_{1,ij}^t, w_{1,ij}, a_{1,ij}$, and $a_{4,ij}$ are the (i,j)th element of the matrices $\mathbf{E}_1^t, \mathbf{W}_1, \mathbf{A}_1$, and $\mathbf{A}_4$.

Then, setting the derivative of (28) to zero, the solution is derived as

$$\frac{\lambda_1}{\eta_1 + \eta_4 w_{1,ij}^2}\operatorname{sign}\left(e_{1,ij}^t\right) + e_{1,ij}^t - \frac{\eta_1 a_{1,ij} - \eta_4 w_{1,ij}a_{4,ij}}{\eta_1 + \eta_4 w_{1,ij}^2} = 0. \tag{29}$$

According to the definition of the soft-thresholding operator $S_\tau(x) = \operatorname{sign}(x)\cdot\max(|x| - \tau, 0)$, the closed-form solution of each element is obtained as

$$e_{1,ij}^t = S_{\frac{\lambda_1}{\eta_1 + \eta_4 w_{1,ij}^2}}\left(\frac{\eta_1 a_{1,ij} - \eta_4 w_{1,ij}a_{4,ij}}{\eta_1 + \eta_4 w_{1,ij}^2}\right). \tag{30}$$

Hence, based on the elementwise soft-thresholding operator, the solution is derived as

$$\mathbf{E}_1^{t,k+1} = S_{\frac{\lambda_1}{\eta_1 + \eta_4\mathbf{W}_1\odot\mathbf{W}_1}}\left(\frac{\eta_1\mathbf{A}_1 - \eta_4\mathbf{W}_1\odot\mathbf{A}_4}{\eta_1 + \eta_4\mathbf{W}_1\odot\mathbf{W}_1}\right). \tag{31}$$

D. Closed-Form Solution of the Subproblem in (22d)

Substituting the augmented Lagrangian function into subproblem (22d), the target echo matrix of channel 2 $\mathbf{E}_2^{t,k+1}$ is updated as (23d), shown at the bottom of the page. Performing variable substitution, the substitution matrices are denoted as $\mathbf{A}_2 = \mathbf{E}_2 - \mathbf{E}_2^{w,k+1} + \mathbf{Z}_2^k/\eta_2, \mathbf{A}_5 = \mathbf{Z}_5^k/\eta_5$.

Similar to (28), the elementwise optimization is constructed

$$\min_{e_{2,ij}^t}\left\{\lambda_2\left|e_{2,ij}^t\right| + \frac{\eta_2}{2}\left(a_{2,ij} - e_{2,ij}^t\right)^2 + \frac{\eta_5}{2}\left(w_{2,ij}e_{2,ij}^t + a_{5,ij}\right)^2\right\} \tag{32}$$

where $e_{2,ij}^t, w_{2,ij}, a_{2,ij}$, and $a_{5,ij}$ are the (i,j)th element of the matrices $\mathbf{E}_2^t, \mathbf{W}_2, \mathbf{A}_2$, and $\mathbf{A}_5$.

$$\begin{aligned}L\left(\mathbf{E}_1^w, \mathbf{E}_2^w, \mathbf{E}_1^t, \mathbf{E}_2^t\right) = {}& \left\|\mathbf{E}_1^w\right\|_* + \left\|\mathbf{E}_2^w\right\|_* + \lambda_1\left\|\mathbf{E}_1^t\right\|_1 + \lambda_2\left\|\mathbf{E}_2^t\right\|_1\\ &+ \frac{\eta_1}{2}\left\|\mathbf{E}_1 - \mathbf{E}_1^w - \mathbf{E}_1^t + \mathbf{Z}_1/\eta_1\right\|_F^2 + \frac{\eta_2}{2}\left\|\mathbf{E}_2 - \mathbf{E}_2^w - \mathbf{E}_2^t + \mathbf{Z}_2/\eta_2\right\|_F^2\\ &+ \frac{\eta_3}{2}\left\|\mathbf{E}_1^w - \mathbf{E}_2^w + \mathbf{Z}_3/\eta_3\right\|_F^2 + \frac{\eta_4}{2}\left\|\mathbf{W}_1\odot\mathbf{E}_1^t + \mathbf{Z}_4/\eta_4\right\|_F^2 + \frac{\eta_5}{2}\left\|\mathbf{W}_2\odot\mathbf{E}_2^t + \mathbf{Z}_5/\eta_5\right\|_F^2\end{aligned} \tag{21}$$

$$\mathbf{E}_1^{w,k+1} = \arg\min_{\mathbf{E}_1^w}\left\{\left\|\mathbf{E}_1^w\right\|_* + \frac{\eta_1}{2}\left\|\mathbf{E}_1 - \mathbf{E}_1^w - \mathbf{E}_1^{t,k} + \mathbf{Z}_1^k/\eta_1\right\|_F^2 + \frac{\eta_3}{2}\left\|\mathbf{E}_1^w - \mathbf{E}_2^{w,k} + \mathbf{Z}_3^k/\eta_3\right\|_F^2\right\} \tag{23a}$$

$$\mathbf{E}_2^{w,k+1} = \arg\min_{\mathbf{E}_2^w}\left\{\left\|\mathbf{E}_2^w\right\|_* + \frac{\eta_2}{2}\left\|\mathbf{E}_2 - \mathbf{E}_2^w - \mathbf{E}_2^{t,k} + \mathbf{Z}_2^k/\eta_2\right\|_F^2 + \frac{\eta_3}{2}\left\|\mathbf{E}_1^{w,k+1} - \mathbf{E}_2^w + \mathbf{Z}_3^k/\eta_3\right\|_F^2\right\} \tag{23b}$$

$$\mathbf{E}_1^{t,k+1} = \arg\min_{\mathbf{E}_1^t}\left\{\lambda_1\left\|\mathbf{E}_1^t\right\|_1 + \frac{\eta_1}{2}\left\|\mathbf{E}_1 - \mathbf{E}_1^{w,k+1} - \mathbf{E}_1^t + \mathbf{Z}_1^k/\eta_1\right\|_F^2 + \frac{\eta_4}{2}\left\|\mathbf{W}_1\odot\mathbf{E}_1^t + \mathbf{Z}_4^k/\eta_4\right\|_F^2\right\} \tag{23c}$$

$$\mathbf{E}_2^{t,k+1} = \arg\min_{\mathbf{E}_2^t}\left\{\lambda_2\left\|\mathbf{E}_2^t\right\|_1 + \frac{\eta_2}{2}\left\|\mathbf{E}_2 - \mathbf{E}_2^{w,k+1} - \mathbf{E}_2^t + \mathbf{Z}_2^k/\eta_2\right\|_F^2 + \frac{\eta_5}{2}\left\|\mathbf{W}_2\odot\mathbf{E}_2^t + \mathbf{Z}_5^k/\eta_5\right\|_F^2\right\} \tag{23d}$$

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Similar to (31), the solution based on the elementwise soft-thresholding operator is derived as

$$\mathbf{E}_2^{t,k+1} = S_{\frac{\lambda_2}{\eta_2+\eta_5\mathbf{W}_2\odot\mathbf{W}_2}}\left(\frac{\eta_2\mathbf{A}_2-\eta_5\mathbf{W}_2\odot\mathbf{A}_5}{\eta_2+\eta_5\mathbf{W}_2\odot\mathbf{W}_2}\right). \tag{33}$$

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